POOJA KIRAN | SECURITY ENGINEERING

Statistical Screening for Poisoned Machine-Learning Training Data

Detection and Quarantine of Suspicious Samples at the Data-Ingestion Boundary

Source inventory: e3d9870f7e095d49b0f4bf61acfb991604bdc028

Test snapshot: 43313ee7dab746a27999e22796074e4a0d941322

CI run 37169444656 | initiated 2026-10-04 UTC | MIT

https://github.com/poojakira/dataset-poisoning-detector

200
tests passed in cited CI

91.20%
statement coverage at cited CI

0.55
centroid F1, synthetic 5% label flips

0.6975
centroid F1, synthetic 20% label flips

CONTRIBUTION

- An ingestion-boundary screening pipeline combining statistical outlier methods with a label-aware cross-class centroid detector that improves the committed synthetic label-flip benchmark from the old top-1 spectral baseline 0.08 / 0.23 / 0.3675 F1 to 0.55 / 0.595 / 0.6975 F1 at 5% / 10% / 20% contamination.

IMPLEMENTED METHOD

- 1. Validate incoming feature batches and streaming samples.
- 2. Apply z-score, IQR, Isolation Forest, and spectral/label-aware detectors.
- 3. Combine detector signals for screening and quarantine workflows.
- 4. Maintain streaming state and deduplication metadata.
- 5. Benchmark label-flip behavior with committed reproducible artifacts.

LIMITATIONS AND EVIDENCE BOUNDARY

- The benchmark is synthetic and separable; it is not proof of production detection rate.
- Clean-label, subtle backdoor, and very-low-rate poisoning remain difficult.
- Throughput is hardware/environment dependent; no fixed product SLA is claimed.
- Stateful streaming behavior requires deliberate partitioning/external state for horizontal scaling.

SCOPED EVIDENCE

- Cited Python 3.12 CI: 200 passed; 91.20% statement coverage, same test job.
- Committed synthetic experiment: 2,000 samples, 100 features, class separation 2.0, seed 2018; not production data.
- At 5/10/20% random class-0-to-class-1 label flips, centroid F1 = 0.55/0.595/0.6975; old top-1 spectral F1 = 0.08/0.23/0.3675; ensemble F1 = 0.0841/0.1421/0.2331.
- Result source: results/spectral_benchmark.json. Recorded artifact values were checked; no new population efficacy estimate is made.

REPRODUCTION

- git checkout 43313ee7dab746a27999e22796074e4a0d941322
- Install the documented development dependencies; run python -m pytest tests/ -q. Use the cited job configuration when reproducing coverage.
- Benchmark data, adapters and environment requirements are separate; consult poster/13_reproducibility.md and the claim ledger.

Dated tests and committed fixtures do not establish universal detection, production deployment or an SLO.